NeuroLink

An educational brain–computer interface simulator

From a noisy waveform to an artificial command

Generate → Filter → Extract → Classify → Visualize

EXPERIENTIAL LEARNING

Team No. 2
Class: CE-SE-A
Biology for Engineers

Educational prototype using simulated EEG-like signals — not live brain data.

Team 2 · contribution & presentation plan

Assigned responsibilities for preparation, explanation and demonstration

MEMBER	STUDENT ID	RESPONSIBILITY	SLIDES
Aditya Dugar	25BRTSN003	Biology, motivation & introduction	1, 3–4
Kushal Chaudhary	25BTRSN025	Acquisition concepts & synthetic signals	5–6
Prince Jatheen Rout	25BTRSN034	Filtering, artifacts & scientific limits	7, 14
Raunak Bothra	25BTRSN037	Features & machine-learning pipeline	8–9
Vijaya Rupesh N	25BTRSN053	Validation, dashboard & live demo	10–12
Yashas L Rao	25BTRSN055	Assistive-tech case study & learning review	13, 15–16

Shared: integration review, rehearsal and questions. Roles are a proposed allocation, not verified individual authorship.

The learning challenge

How does a waveform become a usable command?

Observe

Compare raw and filtered signals using the same time axis.

Measure

Extract two named frequency-band features from each example.

Interpret

Classify generated patterns and explain what the output can—and cannot—mean.

Learning objective: explain each transformation, then test its behavior.

Biology first: what a BCI measures

Neural activity → measurement → signal processing → external output

Real EEG pathway

Electrodes record voltage differences.

Signals vary across people and sessions.

Eye, muscle and motion activity can affect recordings.

Our educational substitution

Python generates a waveform.

Amplitude uses arbitrary units.

No headset, human participant or recorded EEG is involved.

The complete software pipeline

01

Generate

512 samples

02

Filter

1–40 Hz

03

Extract

2 features

04

Classify

left / right

05

Display

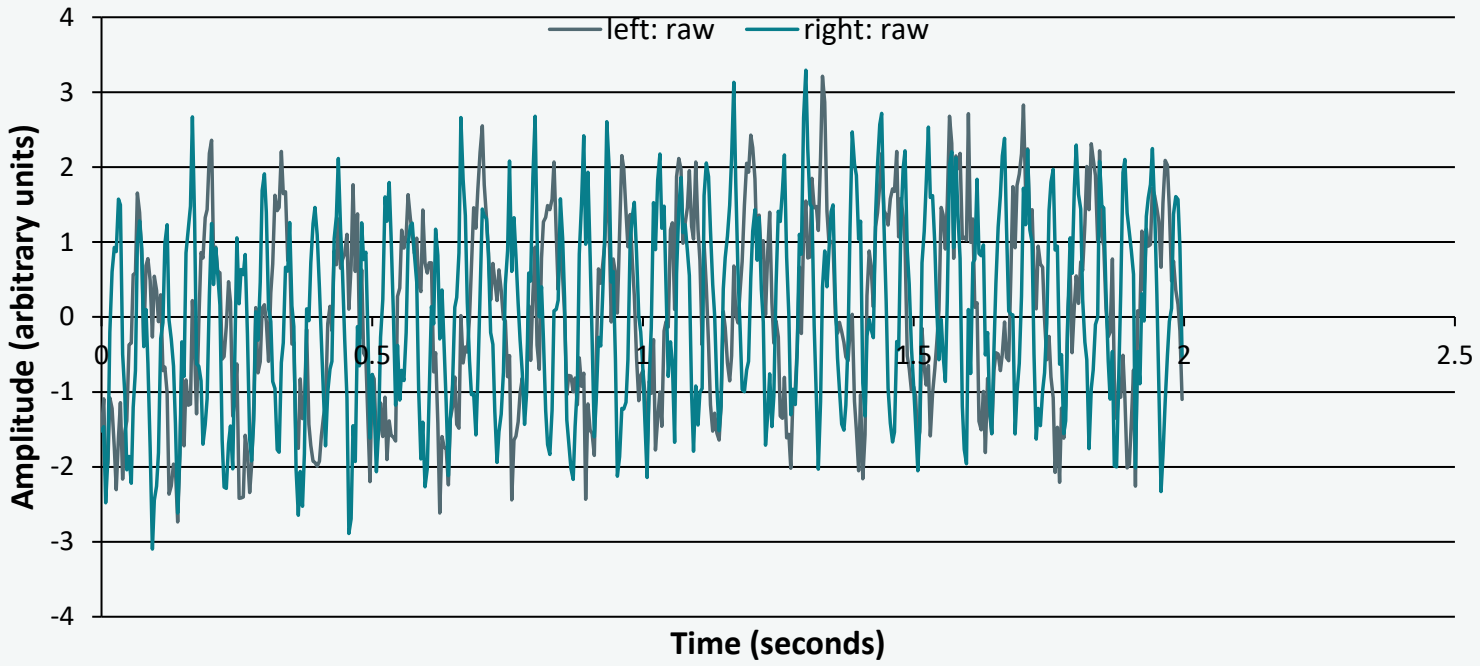
plots + cursor

NumPy · SciPy · scikit-learn · joblib · Streamlit · Plotly

Python 3.12 · Local-first application with an optional Streamlit-hosted demo

Create two deliberately distinct patterns

Actual generated examples · seed 7 · arbitrary amplitudes



256 Hz

2 seconds

512 samples

left: emphasizes 8 Hz

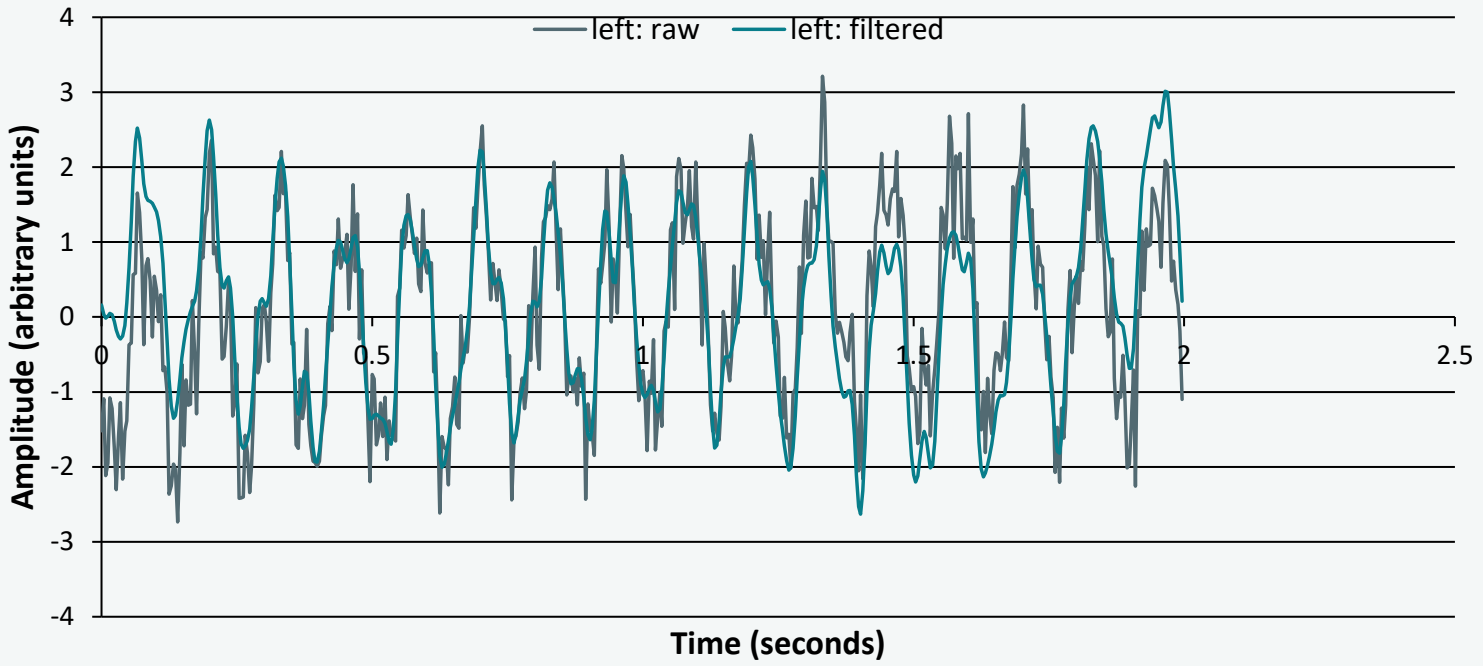
right: emphasizes 22 Hz

Noise + drift + interference

Left and right describe generator settings, not a person’s intention.

Filter the signal, not every artifact

Butterworth band-pass · second-order sections · offline zero phase



Pass band
1–40 Hz

Same sample count

Finite output checked

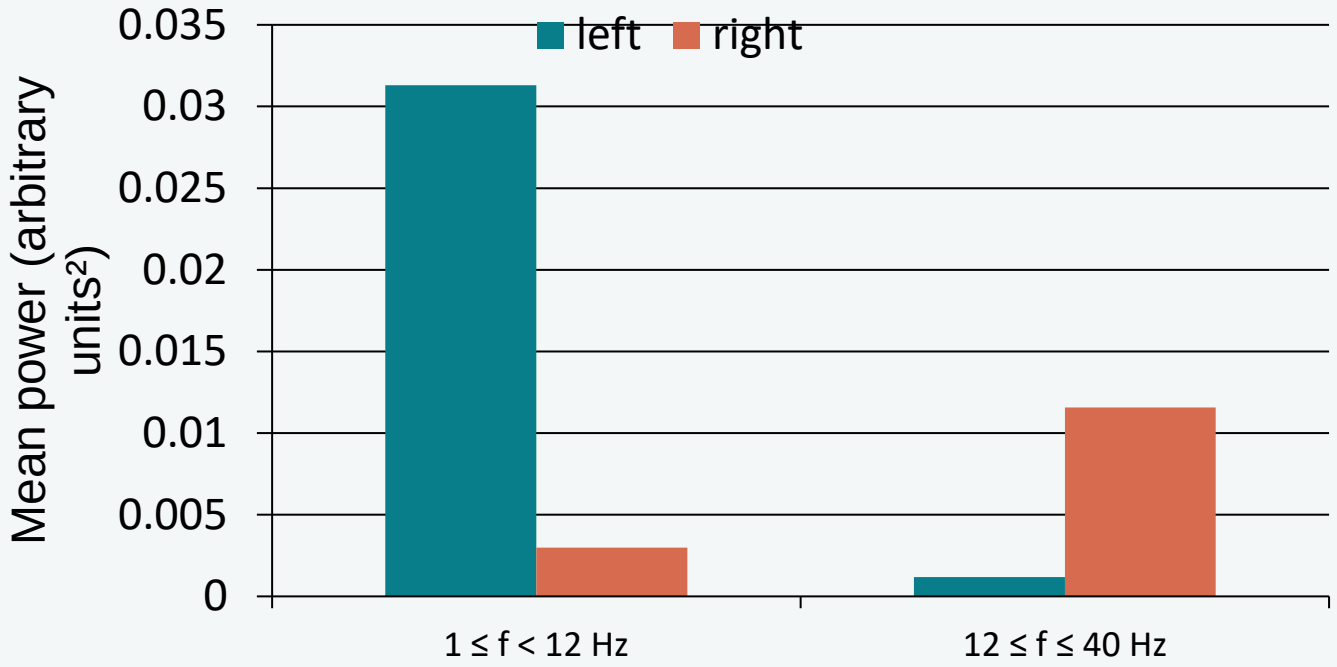
Future samples required

Edge effects remain

Motion, eye and muscle artifacts may overlap the passed frequencies.

Reduce 512 samples to two features

Center → real FFT → squared normalized magnitude → band averages



Stable feature names

mean_power_1_12_hz
mean_power_12_40_hz

Mean $|\mathrm{rFFT}(x - \mathrm{mean}(x)) / N|^2$ within each band

The 12 Hz bin is counted once. Empty bands raise an explicit error.

Train a reproducible classifier

Balanced synthetic data · fixed random state · held-out evaluation

400 examples

200 left + 200 right

Independent example seeds

One generator family

300 / 100 split

Stratified 75% / 25%

Fit scaler on training only

Hold test examples aside

Model pipeline

StandardScaler

LogisticRegression

Save/load with joblib

Reproducibility supports teaching; it does not establish real EEG generalization.

Validation: what the results demonstrate

Source: saved synthetic metrics and locally executed automated tests

100 / 100

held-out synthetic examples correct

58 tests passed locally*

Confusion matrix

Actual → predicted

	left	right
left	50	0
right	0	50

Synthetic accuracy is NOT evidence of real EEG decoding or clinical effectiveness.

*See presenter guide for the verification date and command.

An interactive dashboard makes the stages visible

Interface map · representative values from the actual seed-7 left example

INPUT

Artificial class: left

Seed: 7

Generate and process

Optional simulated cursor

Raw + filtered waveform comparison

Low-band feature: 0.0313

High-band feature: 0.0012

Prediction: left

Model score: 0.9855

Zoomable chart · explicit errors · scientific explanation

Interface schematic; model scores do not measure actual human intention.

Live demonstration · about 90 seconds

Show the causal chain rather than only the final label

01 **Choose left · seed 7**

Compare raw/filtered traces and inspect the low-band feature.

02 **Choose right · seed 7**

Compare the feature balance and predicted artificial label.

03 **Enable cursor · reset to 50**

Left → 40; then right → 50. Bounds are 0–100.

Local backup: python -m streamlit run app.py

Real assistive technology: the BrainGate study

A separate published research example — not a capability of our simulator

Published research

Two participants with tetraplegia

Implanted motor-cortex electrodes

Robotic reaching and grasping

NeuroLink

Generated sinusoidal patterns

Two FFT features and an artificial label

On-screen cursor simulation

Hochberg et al., Nature 485, 372–375 (2012) · doi:10.1038/nature11076

Scientific boundaries we must keep visible

What works in this simulator does not establish what works on a person

Data boundary

Synthetic only

Arbitrary amplitudes

Artificial labels

Method boundary

Offline zero phase

Incomplete artifact removal

Pedagogical band choices

Evidence boundary

Same generator family

No human-intent confidence

No clinical validation

Real EEG is future work requiring a separate dataset, protocol and evaluation.

Experiential learning: build, observe, explain

Connect biology concepts to engineering decisions

Build & inspect

Modular signal processing

Reproducible experiments

Visible intermediate outputs

Test & question

Check shape and finite values

Test frequency behavior

Challenge accuracy claims

Explain & reflect

Identify artifacts

Explain synthetic data

State scientific limits

Discussion: why can 100% synthetic accuracy still tell us very little about real EEG?

References & project resources

Primary research	Hochberg et al. (2012). Reach and grasp by people with tetraplegia using a neurally controlled robotic arm. Nature 485, 372–375. https://pmc.ncbi.nlm.nih.gov/articles/PMC3640850/
Project repository	Source code, README, specifications and setup instructions. https://github.com/CoBrIX44/neurolink-bci-simulator
Methods & evidence	PRD.md · docs/SCIENCE_AND_USER_GUIDE.md · docs/IMPLEMENTATION_NOTES.md · models/neurolink.metrics.json
Software documentation	SciPy signal filtering · scikit-learn pipelines · Streamlit application framework https://docs.scipy.org/doc/scipy/reference/signal.html